

Name: Sarvesh Mishra
Roll No: 2401330100319
Branch: CSE
Section: E
Workshop Lab Number: 012

Question 1

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class JdbcProject {
    private static final String JDBC_URL = "jdbc:mysql://localhost:3306/NIET";
    private static final String USERNAME = "root";
    private static final String PASSWORD = "";

    public static void main(String[] args) {
        try (Connection connection = DriverManager.getConnection(JDBC_URL,
            USERNAME, PASSWORD)) {
            createStudentTable(connection);
            insertSampleStudents(connection);
            displayStudents(connection);
        } catch (SQLException exception) {
            System.out.println("Database operation failed.");
            System.out.println("Reason: " + exception.getMessage());
        }
    }

    private static void createStudentTable(Connection connection) throws
        SQLException {
        String createTableSql = "CREATE TABLE IF NOT EXISTS STUDENT ("
            + "ID INT PRIMARY KEY AUTO_INCREMENT, "
            + "Name VARCHAR(50), "
            + "Age INT, "
            + "Department VARCHAR(50), "
            + "City VARCHAR(50)"
            + ")";

        try (Statement statement = connection.createStatement()) {
            statement.execute(createTableSql);
            System.out.println("STUDENT table created successfully.");
        }
    }

    private static void insertSampleStudents(Connection connection) throws
        SQLException {
        String insertSql = "INSERT INTO STUDENT (Name, Age, Department, City)
            VALUES (?, ?, ?, ?)";

        Object[][] students = {
```

```

        {"Aarav Sharma", 20, "Computer Science", "Noida"},
        {"Ananya Gupta", 21, "Information Technology", "Delhi"},
        {"Rohan Verma", 22, "Electronics", "Ghaziabad"},
        {"Priya Singh", 20, "Mechanical", "Lucknow"},
        {"Kunal Mehta", 23, "Civil", "Jaipur"},
        {"Neha Yadav", 21, "Computer Science", "Gurugram"},
        {"Vivek Kumar", 22, "Information Technology", "Meerut"},
        {"Isha Saxena", 20, "Electronics", "Kanpur"},
        {"Sahil Khan", 24, "Mechanical", "Agra"},
        {"Meera Joshi", 21, "Civil", "Dehradun"}
    };

    try (PreparedStatement preparedStatement =
connection.prepareStatement(insertSql)) {
        for (Object[] student : students) {
            preparedStatement.setString(1, (String) student[0]);
            preparedStatement.setInt(2, (Integer) student[1]);
            preparedStatement.setString(3, (String) student[2]);
            preparedStatement.setString(4, (String) student[3]);
            preparedStatement.addBatch();
        }

        preparedStatement.executeBatch();
        System.out.println("10 sample student records inserted
successfully.");
    }
}

private static void displayStudents(Connection connection) throws SQLException
{
    String selectSql = "SELECT ID, Name, Age, Department, City FROM STUDENT";

    try (Statement statement = connection.createStatement(); ResultSet
resultSet = statement.executeQuery(selectSql)) {
        System.out.println();
        System.out.println("STUDENT RECORDS");
        System.out.printf("%-5s %-20s %-5s %-24s %-15s\n", "ID", "Name",
"Age", "Department", "City");

        while (resultSet.next()) {
            System.out.printf(
                "%-5d %-20s %-5d %-24s %-15s\n",
                resultSet.getInt("ID"),
                resultSet.getString("Name"),
                resultSet.getInt("Age"),
                resultSet.getString("Department"),
                resultSet.getString("City"));
        }
    }
}

```

Output

```
STUDENT table created successfully.  
10 sample student records inserted successfully.
```

STUDENT RECORDS

ID	Name	Age	Department	City
1	Aarav Sharma	20	Computer Science	Noida
2	Ananya Gupta	21	Information Technology	Delhi
3	Rohan Verma	22	Electronics	Ghaziabad
4	Priya Singh	20	Mechanical	Lucknow
5	Kunal Mehta	23	Civil	Jaipur
6	Neha Yadav	21	Computer Science	Gurugram
7	Vivek Kumar	22	Information Technology	Meerut
8	Isha Saxena	20	Electronics	Kanpur
9	Sahil Khan	24	Mechanical	Agra
10	Meera Joshi	21	Civil	Dehradun

Question 2

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;

public class JdbcProjectUpdate {
    private static final String JDBC_URL = "jdbc:mysql://localhost:3306/NIET";
    private static final String USERNAME = "root";
    private static final String PASSWORD = "";

    public static void main(String[] args) {
        try (Connection connection = DriverManager.getConnection(JDBC_URL,
            USERNAME, PASSWORD)) {
            createStudentTable(connection);
            updateStudent(connection);
            deleteStudent(connection);
            insertNewStudents(connection);
            displayStudents(connection);
        } catch (SQLException exception) {
            System.out.println("Database operation failed.");
            System.out.println("Reason: " + exception.getMessage());
        }
    }

    private static void createStudentTable(Connection connection) throws
        SQLException {
        String createTableSql = "CREATE TABLE IF NOT EXISTS STUDENT ("
            + "ID INT PRIMARY KEY AUTO_INCREMENT, "
            + "Name VARCHAR(50), "
            + "Age INT, "
            + "Department VARCHAR(50), "
            + "City VARCHAR(50)"
            + ")";

        try (Statement statement = connection.createStatement()) {
            statement.execute(createTableSql);
        }
    }

    private static void updateStudent(Connection connection) throws SQLException {
        String updateSql = "UPDATE STUDENT SET Age = ?, Department = ? WHERE ID =
?";

        try (PreparedStatement preparedStatement =
            connection.prepareStatement(updateSql)) {
            preparedStatement.setInt(1, 23);
            preparedStatement.setString(2, "Artificial Intelligence");
            preparedStatement.setInt(3, 1);
            int affectedRows = preparedStatement.executeUpdate();
            System.out.println(affectedRows + " student record updated.");
        }
    }

    private static void deleteStudent(Connection connection) throws SQLException {
```

```

        String deleteSql = "DELETE FROM STUDENT WHERE ID = ?";

        try (PreparedStatement preparedStatement =
connection.prepareStatement(deleteSql)) {
            preparedStatement.setInt(1, 2);
            int affectedRows = preparedStatement.executeUpdate();
            System.out.println(affectedRows + " student record deleted.");
        }
    }

    private static void insertNewStudents(Connection connection) throws
SQLException {
        String insertSql = "INSERT INTO STUDENT (Name, Age, Department, City)
VALUES (?, ?, ?, ?)";
        Object[][] students = {
            {"Tanya Bansal", 22, "Data Science", "Noida"},
            {"Aditya Raj", 23, "Cyber Security", "Delhi"}
        };

        try (PreparedStatement preparedStatement =
connection.prepareStatement(insertSql)) {
            for (Object[] student : students) {
                preparedStatement.setString(1, (String) student[0]);
                preparedStatement.setInt(2, (Integer) student[1]);
                preparedStatement.setString(3, (String) student[2]);
                preparedStatement.setString(4, (String) student[3]);
                preparedStatement.addBatch();
            }
            preparedStatement.executeBatch();
            System.out.println("2 new student records inserted.");
        }
    }

    private static void displayStudents(Connection connection) throws SQLException
{
        String selectSql = "SELECT ID, Name, Age, Department, City FROM STUDENT";

        try (Statement statement = connection.createStatement(); ResultSet
resultSet = statement.executeQuery(selectSql)) {
            System.out.println();
            System.out.println("UPDATED STUDENT RECORDS");
            System.out.printf("%-5s %-20s %-5s %-24s %-15s\n", "ID", "Name",
"Age", "Department", "City");

            while (resultSet.next()) {
                System.out.printf(
                    "%-5d %-20s %-5d %-24s %-15s\n",
                    resultSet.getInt("ID"),
                    resultSet.getString("Name"),
                    resultSet.getInt("Age"),
                    resultSet.getString("Department"),
                    resultSet.getString("City"));
            }
        }
    }
}

```

Output

```
1 student record updated.  
1 student record deleted.  
2 new student records inserted.
```

UPDATED STUDENT RECORDS

ID	Name	Age	Department	City
1	Aarav Sharma	23	Artificial Intelligence	Noida
3	Rohan Verma	22	Electronics	Ghaziabad
4	Priya Singh	20	Mechanical	Lucknow
5	Kunal Mehta	23	Civil	Jaipur
6	Neha Yadav	21	Computer Science	Gurugram
7	Vivek Kumar	22	Information Technology	Meerut
8	Isha Saxena	20	Electronics	Kanpur
9	Sahil Khan	24	Mechanical	Agra
10	Meera Joshi	21	Civil	Dehradun
11	Tanya Bansal	22	Data Science	Noida
12	Aditya Raj	23	Cyber Security	Delhi

Question 3

[display.jsp](#)

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%
    RequestDispatcher header = request.getRequestDispatcher("header.jsp");
    header.include(request, response);

    String studentName = (String) request.getAttribute("studentName");
    String rollNumber = (String) request.getAttribute("rollNumber");
%>

<h2>WELCOME TO NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY</h2>

<p>Student Name : <%= studentName %></p>

<p>Roll Number : <%= rollNumber %></p>

<%
    out.flush();
    RequestDispatcher footer = request.getRequestDispatcher("footer.jsp");
    footer.include(request, response);
%>
```

[footer.jsp](#)

```
<p>Thank You for Visiting</p>
```

[header.jsp](#)

```
<h1>NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY</h1>
```

[process.jsp](#)

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%
    String studentName = request.getParameter("studentName");
    String rollNumber = request.getParameter("rollNumber");

    if (studentName == null || studentName.trim().isEmpty()
        || rollNumber == null || rollNumber.trim().isEmpty()) {
        response.sendRedirect("student.jsp");
    } else {
        request.setAttribute("studentName", studentName.trim());
        request.setAttribute("rollNumber", rollNumber.trim());

        RequestDispatcher dispatcher =
request.getRequestDispatcher("display.jsp");
        dispatcher.forward(request, response);
    }
%>
```

[student.jsp](#)

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head>
    <title>Student Information</title>
</head>
<body>
    <h2>Student Information System</h2>

    <form action="process.jsp" method="post">
        <label>Name:</label>
        <input type="text" name="studentName">
        <br><br>

        <label>Roll Number:</label>
        <input type="text" name="rollNumber">
        <br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

[Output](#)

[Student page](#)

Student Information System

Name:

Roll Number:

[Process page](#)

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY

WELCOME TO NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY

Student Name : Rahul Sharma

Roll Number : 101

Thank You for Visiting

Question 4

login.jsp

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<!DOCTYPE html>
<html>
<head>
    <title>Login</title>
</head>
<body>
    <h2>Login Page</h2>

    <form action="validate.jsp" method="post">
        <label>Email:</label>
        <input type="text" name="email">
        <br><br>

        <label>Password:</label>
        <input type="password" name="password">
        <br><br>

        <input type="submit" value="Login">
    </form>
</body>
</html>
```

logout.jsp

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%
    session.invalidate();
%>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="refresh" content="2;url=login.jsp">
    <title>Logged Out</title>
</head>
<body>
    <p>You have successfully logged out.</p>
    <p><a href="login.jsp">Back to login.jsp</a></p>
</body>
</html>
```

[profile.jsp](#)

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%
    String email = (String) session.getAttribute("email");

    if (email == null) {
        response.sendRedirect("login.jsp");
        return;
    }
%>
<!DOCTYPE html>
<html>
<head>
    <title>Profile</title>
</head>
<body>
    <h1>WELCOME NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY</h1>
    <p>Logged in as: <%= email %></p>
    <p><a href="logout.jsp">Logout</a></p>
</body>
</html>
```

[validate.jsp](#)

```
<%@ page contentType="text/html; charset=UTF-8" language="java" %>
<%
    String email = request.getParameter("email");
    String password = request.getParameter("password");

    if ("admin@niet.co.in".equals(email) && "12345".equals(password)) {
        session.setAttribute("email", email);
        response.sendRedirect("profile.jsp");
        return;
    }
%>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="refresh" content="2;url=login.jsp">
    <title>Invalid Login</title>
</head>
<body>
    <p>Invalid email or password. Please try again.</p>
    <p><a href="login.jsp">Back to login.jsp</a></p>
</body>
</html>
```

[Output](#)

[Login page](#)

Login Page

Email:

Password:

Login

[Profile page](#)

WELCOME NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY

Logged in as: admin@niet.co.in

[Logout](#)

[Logout page](#)

Invalid email or password. Please try again.

[Back to login.jsp](#)